

Luke Thompson

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EDUCATION

The University of Sydney

PhD, Computer Science

Sydney, AU

Jun 2024 — Present

- Working on AI for science using neural operators (PDE solvers) and rough path theory (stochastic calculus)
- Awarded a AUD 75 000 CSIRO PhD scholarship to continue my work on graph transformers (declined)

The University of Sydney

Bachelor of Science (Honours class I), Pharmacology

Sydney, AU

Feb 2020 — Jun 2024

- Average Mark: 86 | First Class Honours, Dalyell Scholar (declined)
- **Thesis:** Graph Transformer Neural Networks for Mutagenicity Prediction

PUBLICATIONS

Learning Manifold and Ito Dynamics with Branched Neural Rough Differential Equations

September 2025

Luke Thompson, Dai Shi, Lequan Lin, Andi Han, Junbin Gao

Submitted ICML2026

- Developed a model employing formalisms from abstract algebra to define stochastic processes over manifolds
- Built a large open source Jax library to facilitate this work

Expanding the Chaos: A Neural Operator for Stochastic (Partial) Differential Equations

September 2025

Luke Thompson, Dai Shi, Lequan Lin, Andi Han, Junbin Gao

Submitted ICML2026

- Led parameter estimation, stochastic volatility, and rough path components of the paper
- Contributed code for stochastic differential equations on Riemannian manifolds

ATOM: A Pretrained Neural Operator for Molecular Dynamics

September 2025

Luke Thompson, Davy Guan, Dai Shi, Slade Matthews, Junbin Gao, Andi Han

ICLR2026

- State-of-the-Art molecular dynamics forecasting with an efficiently trainable transformer architecture
- Developed a custom form of RoPE for dynamics learning, leveraged equivariant deep learning methods
- Worked with the CSIRO to build a multitask molecular dataset using HPC resources (total cost US 130 000)

AmesFormer: State-of-the-Art Mutagenicity Prediction with Graph Transformers

September 2024

Luke Thompson, Josiah Evans, Slade Matthews

Chemical Research in Toxicology

- First of 26 models for carcinogenicity prediction from chemical structure
- Implemented using PyTorch, PyTorch-Geometric, and a custom Rust library

WORK EXPERIENCE

Casual Academic

June 2024 — Present

The University of Sydney

Sydney, Australia

- Tutored third year and master's business analytics. Covered Bayesian machine learning, Markov chain Monte Carlo, and foundational methods
- Led two capstone project groups focusing on computational chemistry

Geographic Data Analyst

Feb 2024 — Present

Kumon

Sydney, Australia

- Saved AUD 28 000 per annum by bringing population time-series forecasting in-house using an auto-regressive ensemble
- Migrated four legacy Australia-wide Visual Basic systems to a high-performance Rust and PostgreSQL back end, reducing latency from approximately four hours to 30 seconds for common tasks and achieving 99% uptime

PROJECTS

Stochastax

<https://github.com/luke-a-thompson/QuickSig>

- A Jax library for rough path computations
- Covers path signatures, abstract algebra, log signatures, rough differential equations, and several miscellaneous features
- Interoperates with a Rust computer algebra system for abstract algebra computations

FastASX

<https://github.com/luke-a-thompson/FastASX>

- Fastest open-source NASDAQ ITCH message parser. Zero-allocation, parallel, built with Rust
- Maintains a limit order book and intraday stock directory

Cold War: Iron Curtain

<https://github.com/swf541/ColdWarIronCurtain>

- Founded an open-source strategy game modification, achieving 600 000 users before I passed it on to other maintainers

AWARDS AND PRESENTATIONS

London Institutde of Mathematical Sciences Stochastic PDE Prize

2025

- My team placed first in a contest focused on training a machine learning model predict the evolution the φ_2^4 equation, which describes a scalar quantum field. Our winning method was a Polynomial chaos neural operator

ACTRA Student Prize

2024

- I was awarded the student prize for my paper AmesFormer. Presented at the annual scientific meeting of the Australasian College of Toxicologists. Valued at AUD 2000

Sydney Business Scholar Brown Bag Series: Presented my progress on developing operator learning for rough paths

Federal Department of Climate Change: Presented on use of graph neural networks for ecosystem impact prediction

AICIS, federal chemical regulator: Presented AmesFormer to a panel of regulatory toxicologists

SKILLS

- **Programming Languages:** Rust, Python (PyTorch, Polars, Jax, Torch-Geometric, sklearn, Pandas), MATLAB, PostgreSQL, LaTeX
- **Technologies:** Office Suite, Dragon, and related cheminformatics software